

Avi Tripathi



Design Engineer | UX, ML and Full-Stack Dev

01 EDUCATION

Sept 2020 — Jun 2024

Imperial College London, Dyson School

MEng Design Engineering

- **Key Modules:** Human-Centered Design, Machine Learning, Enterprise Roll-out, Innovation and Entrepreneurship, UX Design
- **Online Design Portfolio:** avitripathi.com

02 EXPERIENCE

09/2024 — Present

ML Engineer/UX Designer at Autodesk

- Conducted **Automation Research** for the **Innovation Team** for Fusion Manufacturing
- Integrated custom encoder ML models into **full-stack prototypes using figma, react, typescript, node.js** - novel ML solution released to **15,000+ insider customers**.
- Assisted in building the **largest manufacturing dataset in the industry, 10+million projects**, using distributed data processing pipelines with ray, sagemaker and metaflow.
- Presented ML and UX talks to 100s of developers at conferences.

09/2023 — 09/2024

Full Stack Developer at Setmixer (start-up)

- Developed end-to-end **Figma UX/UI Mockups** for music streaming website, desktop and mobile. Supported in development of **React/Typescript/Node** app, now released to users - <https://listen.setmixer.com/>
- Led development of advanced live musical instrument classification ML model and live music data pipeline, achieving **3% performance improvement over state-of-the-art**, using CNNs, transfer learning and signal processing.

09/2023 — 07/2024

Founding XD Engineer at Patent Space (start-up)

- Built an infringement risk management start-up from scratch which was a **finalist of the London Mayor's Entrepreneur Competition 2024**.
- Led development of **UX and front end prototypes using Figma**, for an **AI enriched patent search and infringement monitoring tool**, backed by user research and interviews.

08/2022 — 08/2024

Brand Manager at ICL Design Engineering Society

- Maintained the brand for a society of **over 500** students through design of media and merchandise, whilst also leading advertising and supply, achieving **over £3000** in sales.
- Individually created the visual identity for **2 Open House events** held with the Great Exhibition Road Festival, attracting over **50,000 visitors**.

03 KEY SKILLS

XD | Figma (3 years)

Graphics | Ps, Ai (5 years)

Motion | Ae, Blender (3 years)

ML Models | LLMs, Encoders, Pytorch

WebDev | React, Typescript

Scripting | Python, C++